Workspace lab

In this lab, you'll learn how to create, switch, and use workspaces in the msfconsole. Workspaces in the very simplest term is basically a save state. When you're trying to find out what ports and protocol is open on a network. You wouldn't want to start over have to carry out reconnaissance every time. You want to keep your information stored. That's where workspaces come in. You can Nmap a network and find out what ports is open and what services is running on the ports and record it using your workspace.

Step 1. Open up your Kali Linux VM and open the terminal so you can make yourself root user.

```
┌──(kali㉿kali)-[~]
└─$ sudo su
[sudo] password for kali:
┌──(root㉿kali)-[/home/kali]
└─#
```

Step 2. Start the postgresql by using the command **"service postgresql start".** Then you can check if it's actually running by using the command **"service postgresql status"**. If you're unable to input anything after checking the status, press "Ctrl + c"

```
┌──(root㉿kali)-[/home/kali]
└─# service postgresql start

┌──(root㉿kali)-[/home/kali]
└─# service postgresql status
● postgresql.service - PostgreSQL RDBMS
     Loaded: loaded (/lib/systemd/system/postgresql.service; disabled; vendor preset: disabled)
     Active: active (exited) since Fri 2022-09-09 22:02:28 EDT; 45s ago
    Process: 1588 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 1588 (code=exited, status=0/SUCCESS)
        CPU: 858us

Sep 09 22:02:28 kali systemd[1]: Starting PostgreSQL RDBMS ...
Sep 09 22:02:28 kali systemd[1]: Finished PostgreSQL RDBMS.
```

Step 3. Afterward you want to initialize the msf database by typing **“msfdb init”**.

```
┌──(root㉿kali)-[/home/kali]
└─# msfdb init
[+] Starting database
```

Step 4. Enter your msfconsole. You can do so by using the command I gave in the previous lab or you can do **“msfconsole -q”** the -q removes the banner and launches msfconsole in quiet mode.

```
┌──(root㉿kali)-[/home/kali]
└─# msfconsole -q
```

Step 5. Now you’ll want to check all the workspaces you have by typing the command **“workspace -v”.** This will displace all the workspace and their host, services, and etc. But if you just want to see which workspace, you’re on you can just type **“workspace”** This will display less info.

```
msf6 > workspace -v

Workspaces
==========

current  name     hosts  services  vulns  creds  loots  notes
-------  ----     -----  --------  -----  -----  -----  -----
*        default  0      0         0      0      0      0
```

Step 6. I only have 1 workspace so I’ll be creating another one using the command **“workspace -a meta3”** meta3 is just purely the name you use for this workspace so you can replace it with another name if you so feel. Then use the **“workspace -v”** command to see that you created it.

```
msf6 > workspace -a meta3
[*] Added workspace: meta3
[*] Workspace: meta3
msf6 > workspace -v

Workspaces
==========

current  name     hosts  se
-------  ----     -----  --
         default  0      0
*        meta3    0      0
```

Step 7. Now that you've created workspace meta3 we can Nmap our Metasploitable 3 vm and find out what port and services is running on it. So launch your Meta 3 machine. You'll also want to find the IP address because you're going to need it to Nmap the machine. Make sure the Kali and Metasploitable 3 is on the same Nat Network. For me my IPv4 is 10.0.2.20. Yours might differ.

```
Ethernet adapter Local Area Connection:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::ec25:
   IPv4 Address. . . . . . . . . . . : 10.0.2.20
```

Step 8. Now you've done all the prep and you're ready to Nmap Meta3. To do so use the command **"db_nmap -p- -sV -O 10.0.2.20"** -p- is letting the nmap know to check every single port because we don't know which one is open. -sV is to check what services is actually running on reach port because it's possible to change which protocol actually run on each services instead of using the default. Finally -O is to check the operating system that the machine is running. This can take up to 5 minutes, so I recommend getting coffee, water, or whatever you want.

```
msf6 > db_nmap -p- -sV -O 10.0.2.20
[*] Nmap: Starting Nmap 7.92 ( https://nmap.org ) at 2022-09-09 22:37 EDT
```

Step 9. When the nmap scan finish you can type **"workspace -v"** to see that the amount of services has been updated.

```
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 290.51 seconds
msf6 > workspace -v

Workspaces
==========

current  name     hosts  services  vulns  creds  loots  notes
-------  ----     -----  --------  -----  -----  -----  -----
         default  0      0         0      0      0      0
*        meta3    1      19        0      0      0      2
```

Step 10. If you want to see which ports is open and the protocol running on that port you have to type **"services"**

```
host       port   proto  name            state  info
----       ----   -----  ----            -----  ----
10.0.2.20  21     tcp    ftp             open   Microsoft ftpd
10.0.2.20  22     tcp    ssh             open   OpenSSH 7.1 protocol 2.0
10.0.2.20  80     tcp    http            open   Microsoft IIS httpd 7.5
10.0.2.20  1617   tcp    java-rmi        open   Java RMI
10.0.2.20  4848   tcp    ssl/http        open   Oracle Glassfish Application Server
10.0.2.20  5985   tcp    http            open   Microsoft HTTPAPI httpd 2.0 SSDP/UPnP
10.0.2.20  8020   tcp    http            open   Apache httpd
10.0.2.20  8022   tcp    http            open   Apache Tomcat/Coyote JSP engine 1.1
10.0.2.20  8027   tcp    papachi-p2p-srv open
10.0.2.20  8080   tcp    http            open   Sun GlassFish Open Source Edition  4.0
10.0.2.20  8282   tcp    http            open   Apache Tomcat/Coyote JSP engine 1.1
10.0.2.20  8383   tcp    http            open   Apache httpd
10.0.2.20  8484   tcp    http            open   Jetty winstone-2.8
10.0.2.20  8585   tcp    http            open   Apache httpd 2.2.21 (Win64) PHP/5.3.10 DAV/2
10.0.2.20  9200   tcp    wap-wsp         open
10.0.2.20  49153  tcp    msrpc           open   Microsoft Windows RPC
10.0.2.20  49154  tcp    msrpc           open   Microsoft Windows RPC
10.0.2.20  49159  tcp    java-rmi        open   Java RMI
10.0.2.20  49160  tcp    tcpwrapped      open

msf6 >
```

Now you're done with this lab. I want you to think about how this can be helpful in gaining access to the victim's machine and what be one of your first steps to try and exploit the victim machine using this info.